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Camera lens group

Abstract

The present disclosure discloses a camera lens group including, sequentially from an object side to an image side along an optical axis, a stop; a first lens having positive refractive power and a convex object-side surface; a second lens having refractive power; a third lens having refractive power, a convex object-side surface and a concave image-side surface; a fourth lens having positive refractive power, a concave object-side surface and a convex image-side surface; and a fifth lens having negative refractive power, a convex object-side surface and a concave image-side surface.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8233224	12/2011	Chen	359/764	G02B 9/60
8482863	12/2012	Tsai	359/767	G02B 13/18
2010/0253829	12/2009	Shinohara	359/764	G02B 13/0045
2011/0013069	12/2010	Chen	348/335	G02B 9/60
2011/0043930	12/2010	Kwon	N/A	N/A
2012/0069455	12/2011	Lin	359/714	G02B 13/0045
2012/0154929	12/2011	Tsai	359/714	G02B 9/60
2013/0002920	12/2012	Sano	N/A	N/A
2013/0016261	12/2012	Tanaka	359/708	G02B 13/0045
2013/0279023	12/2012	Chen	N/A	N/A
2014/0063620	12/2013	Jung et al.	N/A	N/A
2014/0293445	12/2013	Yoneyama	359/714	G02B 13/0045
2014/0340766	12/2013	Hsu	359/714	G02B 13/0045
2015/0103225	12/2014	Hsu	348/335	G02B 9/60
2015/0253537	12/2014	Ye	348/374	G02B 9/60
2018/0074293	12/2017	Hsu	N/A	G02B 5/208

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101995641	12/2010	CN	N/A
102317834	12/2011	CN	N/A
102540423	12/2011	CN	N/A
102566016	12/2011	CN	N/A
102566018	12/2011	CN	N/A
102681142	12/2011	CN	N/A
102955223	12/2012	CN	N/A
102959448	12/2012	CN	N/A
102959449	12/2012	CN	N/A
103048774	12/2012	CN	N/A
103389568	12/2012	CN	N/A
103472568	12/2012	CN	N/A

103676095	12/2013	CN	N/A
103869453	12/2013	CN	N/A
104238080	12/2013	CN	N/A
104914558	12/2014	CN	N/A
105866933	12/2015	CN	N/A
109960010	12/2018	CN	N/A
110261994	12/2018	CN	N/A
110376708	12/2018	CN	N/A
110596865	12/2018	CN	N/A
110727083	12/2019	CN	N/A
111221106	12/2019	CN	N/A
111239971	12/2019	CN	N/A
210720853	12/2019	CN	N/A
2020-109491	12/2019	CN	N/A
211528805	12/2019	CN	N/A
211554456	12/2019	CN	N/A
211577551	12/2019	CN	N/A
211786312	12/2019	CN	N/A
211786341	12/2019	CN	N/A
212905663	12/2020	CN	N/A
2012208148	12/2011	JP	N/A
20120094729	12/2011	KR	N/A
101552594	12/2014	KR	N/A
20160117998	12/2015	KR	N/A
20160130972	12/2015	KR	N/A
I421557	12/2013	TW	N/A
201409110	12/2013	TW	N/A
201939085	12/2018	TW	N/A
I674450	12/2018	TW	N/A
202016601	12/2019	TW	N/A
2013058111	12/2012	WO	N/A

OTHER PUBLICATIONS

First Examination Report for Application No. 202114011155, dated Feb. 2, 2022, 4 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims benefit of priority to Chinese Patent Application No. 202010522818.5 filed on Jun. 10, 2020 before the China National Intellectual Property Administration, the entire disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of optical elements, and specifically, relates to a

camera lens group.

BACKGROUND

(3) With the continuous improvement of related technologies of portable electronic devices, such as smart phones, the trend of using portable electronic devices, such as smart phones, to take pictures instead of traditional cameras has become more and more obvious. At present, most of the mainstream models of portable electronic devices, such as smart phones, on the market are mainly full-screen models. The screen ratio of portable electronic devices, such as smart phones, is getting higher and higher, which will make the front cameras suitable for traditional mobile phones and other electronic devices no longer applicable.

(4) Under this trend, lens assemblies with small front end have won the favor of developers of portable electronic devices, such as smartphones, with their extremely small front end size. However, the traditional center-mounted wide-angle lens assembly cannot achieve the characteristics of extremely small front end. How to make the camera lens group have an ultra-small front end on the basis of achieving a wide angle is one of the problems that many lens assembly designers need to solve urgently. The camera lens group with ultra-small front end and wide angle is more suitable for portable electronic devices, such as full-screen smartphones.

SUMMARY

(5) The present disclosure provides a camera lens group which includes, sequentially from an object side to an image side along an optical axis, a stop; a first lens having positive refractive power and a convex object-side surface; a second lens having refractive power; a third lens having refractive power, a convex object-side surface and a concave image-side surface; a fourth lens having positive refractive power, a concave object-side surface and a convex image-side surface; and a fifth lens having negative refractive power, a convex object-side surface and a concave image-side surface. Half of a maximum field-of-view Semi-FOV of the camera lens group and a total effective focal length f of the camera lens group may satisfy: $2.00 \text{ mm} < \tan.\sup.2(\text{Semi-FOV}) * f < 5.00 \text{ mm}$ A distance SAG42 along the optical axis from an intersection of the image-side surface of the fourth lens and the optical axis to a vertex of an effective radius of the image-side surface of the fourth lens and a distance SAG51 along the optical axis from an intersection of the object-side surface of the fifth lens and the optical axis to a vertex of an effective radius of the object-side surface of the fifth lens may satisfy: $1.00 < (\text{SAG42} + \text{SAG51}) / (\text{SAG42} - \text{SAG51}) < 3.50$.

(6) In one embodiment, at least one of the object-side surface of the first lens to the image-side surface of the fifth lens is aspheric.

(7) In one embodiment, a maximum effective radius DT11 of the object-side surface of the first lens and a maximum effective radius DT52 of the image-side surface of the fifth lens may satisfy: $3.00 < \text{DT52} / \text{DT11} < 5.00$.

(8) In one embodiment, a distance TTL along the optical axis from the object-side surface of the first lens to an imaging plane of the camera lens group and half of a diagonal length ImgH of an effective pixel area on the imaging plane of the camera lens group may satisfy: $\text{TTL} / \text{ImgH} < 1.55$.

(9) In one embodiment, an effective focal length $f1$ of the first lens and a radius of curvature R1 of the object-side surface of the first lens may satisfy: $1.00 < f1 / R1 < 6.00$.

(10) In one embodiment, a radius of curvature R5 of the object-side surface of the third lens and a radius of curvature R10 of the image-side surface of the fifth lens may satisfy: $2.00 < R5 / R10 < 6.00$.

(11) In one embodiment, a center thickness CT4 of the fourth lens along the optical axis and a spaced interval T34 between the third lens and the fourth lens along the optical axis may satisfy: $1.00 < \text{CT4} / \text{T34} < 4.00$.

(12) In one embodiment, an edge thickness ET4 of the fourth lens and an edge thickness ET5 of the fifth lens may satisfy: $1.00 < (\text{ET4} + \text{ET5}) / (\text{ET5} - \text{ET4}) < 4.00$.

(13) In one embodiment, a center thickness CT5 of the fifth lens along the optical axis and a center thickness CT3 of the third lens along the optical axis may satisfy: $0.50 < \text{CT5} / \text{CT3} < 2.50$.

(14) In one embodiment, the total effective focal length f of the camera lens group and an effective

focal length f_4 of the fourth lens may satisfy: $1.00 < f/f_4 < 3.00$.

(15) In another aspect, the present disclosure provides a camera lens group which includes, sequentially from an object side to an image side along an optical axis, a stop; a first lens having positive refractive power and a convex object-side surface; a second lens having refractive power; a third lens having refractive power, a convex object-side surface and a concave image-side surface; a fourth lens having positive refractive power, a concave object-side surface and a convex image-side surface; and a fifth lens having negative refractive power, a convex object-side surface and a concave image-side surface. Half of a maximum field-of-view Semi-FOV of the camera lens group and a total effective focal length f of the camera lens group may satisfy: $2.00 \text{ mm} < \tan.\sup.2(\text{Semi-FOV}) * f < 5.00 \text{ mm}$. An edge thickness ET_4 of the fourth lens and an edge thickness ET_5 of the fifth lens may satisfy: $1.00 < (ET_4 + ET_5) / (ET_5 - ET_4) < 4.00$.

(16) The present disclosure employs a plurality of (for example, five) lenses, and the above camera lens group has at least one beneficial effect, such as miniaturization, wide-angle, ultra-small front end, and high image quality and the like, by rationally configuring the refractive power, the surface shape, the center thickness of each lens, and the on-axis spaced interval between the lenses and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other features, objects, and advantages of the present disclosure will become more apparent by reading the detailed description of the non-limiting embodiments with reference to the accompanying drawings:

(2) FIG. 1 illustrates a schematic structural view of a camera lens group according to example 1 of the present disclosure;

(3) FIGS. 2A to 2D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 1, respectively;

(4) FIG. 3 illustrates a schematic structural view of a camera lens group according to example 2 of the present disclosure;

(5) FIGS. 4A to 4D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 2, respectively;

(6) FIG. 5 illustrates a schematic structural view of a camera lens group according to example 3 of the present disclosure;

(7) FIGS. 6A to 6D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 3, respectively;

(8) FIG. 7 illustrates a schematic structural view of a camera lens group according to example 4 of the present disclosure;

(9) FIGS. 8A to 8D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 4, respectively;

(10) FIG. 9 illustrates a schematic structural view of a camera lens group according to example 5 of the present disclosure;

(11) FIGS. 10A to 10D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 5, respectively;

(12) FIG. 11 illustrates a schematic structural view of a camera lens group according to example 6 of the present disclosure;

(13) FIGS. 12A to 12D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 6, respectively;

(14) FIG. 13 illustrates a schematic structural view of a camera lens group according to example 7 of the present disclosure;

- (15) FIGS. 14A to 14D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 7, respectively;
- (16) FIG. 15 illustrates a schematic structural view of a camera lens group according to example 8 of the present disclosure;
- (17) FIGS. 16A to 16D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 8, respectively;
- (18) FIG. 17 illustrates a schematic structural view of a camera lens group according to example 9 of the present disclosure; and
- (19) FIGS. 18A to 18D illustrate longitudinal aberration curves, astigmatic curves, a distortion curve, and a lateral color curve of the camera lens group of the example 9, respectively.

DETAILED DESCRIPTION OF EMBODIMENTS

- (20) For a better understanding of the present disclosure, various aspects of the present disclosure will be described in more detail with reference to the accompanying drawings. It should be understood that the detailed description is merely illustrative of the exemplary embodiments of the present disclosure and is not intended to limit the scope of the present disclosure in any way. Throughout the specification, the same reference numerals refer to the same elements. The expression “and/or” includes any and all combinations of one or more of the associated listed items.
- (21) It should be noted that in the present specification, the expressions such as first, second, third are used merely for distinguishing one feature from another, without indicating any limitation on the features. Thus, a first lens discussed below may also be referred to as a second lens or a third lens without departing from the teachings of the present disclosure.
- (22) In the accompanying drawings, the thickness, size and shape of the lens have been somewhat exaggerated for the convenience of explanation. In particular, shapes of spherical surfaces or aspheric surfaces shown in the accompanying drawings are shown by way of example. That is, shapes of the spherical surfaces or the aspheric surfaces are not limited to the shapes of the spherical surfaces or the aspheric surfaces shown in the accompanying drawings. The accompanying drawings are merely illustrative and not strictly drawn to scale.
- (23) Herein, the paraxial area refers to an area near the optical axis. If a surface of a lens is convex and the position of the convex is not defined, it indicates that the surface of the lens is convex at least in the paraxial region; and if a surface of a lens is concave and the position of the concave is not defined, it indicates that the surface of the lens is concave at least in the paraxial region. In each lens, the surface closest to the object is referred to as an object-side surface of the lens, and the surface closest to the imaging plane is referred to as an image-side surface of the lens.
- (24) It should be further understood that the terms “comprising,” “including,” “having,” “containing” and/or “contain,” when used in the specification, specify the presence of stated features, elements and/or components, but do not exclude the presence or addition of one or more other features, elements, components and/or combinations thereof. In addition, expressions, such as “at least one of,” when preceding a list of features, modify the entire list of features rather than an individual element in the list. Further, the use of “may,” when describing embodiments of the present disclosure, refers to “one or more embodiments of the present disclosure.” Also, the term “exemplary” is intended to refer to an example or illustration.
- (25) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by those of ordinary skill in the art to which the present disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with the meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.
- (26) It should also be noted that, the examples in the present disclosure and the features in the examples may be combined with each other on a non-conflict basis. The present disclosure will be

described in detail below with reference to the accompanying drawings and in combination with the examples.

(27) The features, principles, and other aspects of the present disclosure are described in detail below.

(28) A camera lens group according to an exemplary embodiment of the present disclosure may include five lenses having refractive power, which are a first lens, a second lens, a third lens, a fourth lens and a fifth lens. The five lenses are arranged sequentially from an object side to an image side along an optical axis. Among the first lens to the fifth lens, there may be a spaced interval between each two adjacent lenses.

(29) In an exemplary embodiment, the first lens may have positive refractive power, and an object-side surface thereof may be convex; the second lens has positive or negative refractive power; the third lens has positive or negative refractive power, an object-side surface thereof may be convex, and an image-side surface thereof may be concave; the fourth lens may have positive refractive power, an object-side surface thereof may be concave, and an image-side surface thereof may be convex; and the fifth lens may have negative refractive power, an object-side surface thereof may be convex, and an image-side surface thereof may be concave.

(30) When cooperates the first lens having positive refractive power with the fourth lens having positive refractive power and the fifth lens having negative refractive power, the camera lens group may maintain better correction ability for chromatic aberration while expanding the maximum field-of-view of the camera lens group, thereby having a better image quality. By designing the object-side surface of the first lens as a convex surface, it may be beneficial to reduce the sensitivity of the camera lens group, thereby theoretically improving the mass production yield of the camera lens group as much as possible. By designing the object-side surface of the third lens as a convex surface and the image-side surface of the third lens as a concave surface, and cooperating with the concave-convex shape of the fourth lens, it is beneficial to reduce the risk of ghosting caused by the reflection between the third lens and the fourth lens and the internal reflection of the fourth lens as much as possible without reducing the image quality of the camera lens group. By designing the fifth lens as a convex-concave shape lens, it is beneficial to improve the image quality in the center imaging area, so that the camera lens group has a better image quality.

(31) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $2.00 \text{ mm} < \tan^2(\text{Semi-FOV}) \cdot f < 5.00 \text{ mm}$, where Semi-FOV is half of a maximum field-of-view of the camera lens group, and f is a total effective focal length of the camera lens group. More specifically, Semi-FOV and f may further satisfy: $2.70 \text{ mm} < \tan^2(\text{Semi-FOV}) \cdot f < 4.30 \text{ mm}$. When $2.00 \text{ mm} < \tan^2(\text{Semi-FOV}) \cdot f < 5.00 \text{ mm}$ is satisfied, the field-of-view of the camera lens group may be increased, and the image height of the image may be controlled within a reasonable range, so that the lens assembly may have better imaging capabilities.

(32) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $3.00 < DT52/DT11 < 5.00$, where DT11 is a maximum effective radius of the object-side surface of the first lens, and DT52 is a maximum effective radius of the image-side surface of the fifth lens. More specifically, DT52 and DT11 may further satisfy: $3.20 < DT52/DT11 < 4.80$. Satisfying $3.00 < DT52/DT11 < 5.00$ helps to the design of the depth of the lens barrel, making the lens assembly more suitable for the front camera.

(33) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $TTL/ImgH < 1.55$, where TTL is a distance along the optical axis from the object-side surface of the first lens to an imaging plane of the camera lens group, and ImgH is half of a diagonal length of an effective pixel area on the imaging plane of the camera lens group. Satisfying $TTL/ImgH < 1.55$ may avoid the problem of the overall size of the camera lens group being too long due to the excessive thickness of the first lens while ensuring the basic imaging height, which is beneficial to maintain the ultra-small characteristics of the camera lens group.

(34) In an exemplary embodiment, the camera lens group according to the present disclosure may

satisfy: $1.00 < f_1/R_1 < 6.00$, where f_1 is an effective focal length of the first lens, and R_1 is a radius of curvature of the object-side surface of the first lens. More specifically, f_1 and R_1 may further satisfy: $1.30 < f_1/R_1 < 5.80$. Satisfying $1.00 < f_1/R_1 < 6.00$ may increase the field-of-view of the camera lens group while avoiding the risk of the first lens being too sensitive and not conducive to processing due to too small f_1 or too large R_1 .

(35) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $2.00 < R_5/R_{10} < 6.00$, where R_5 is a radius of curvature of the object-side surface of the third lens, and R_{10} is a radius of curvature of the image-side surface of the fifth lens. More specifically, R_5 and R_{10} may further satisfy: $2.20 < R_5/R_{10} < 6.00$. When $2.00 < R_5/R_{10} < 6.00$ is satisfied, the correction ability of the camera lens group for chromatic aberration and distortion may be guaranteed, and specifically, the image quality in the central area may be improved advantageously.

(36) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $1.00 < CT_4/T_{34} < 4.00$, where CT_4 is a center thickness of the fourth lens along the optical axis, and T_{34} is a spaced interval between the third lens and the fourth lens along the optical axis. More specifically, CT_4 and T_{34} may further satisfy: $1.30 < CT_4/T_{34} < 3.20$. Satisfying $1.00 < CT_4/T_{34} < 4.00$ is beneficial to reduce the risk of ghost images generated by the internal reflection of the fourth lens and the reflection between the fourth lens and the third lens, and thus the image quality of the camera lens group is advantageously improved.

(37) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $1.00 < (SAG_{42} + SAG_{51}) / (SAG_{42} - SAG_{51}) < 3.50$, where SAG_{42} is a distance along the optical axis from an intersection of the image-side surface of the fourth lens and the optical axis to a vertex of an effective radius of the image-side surface of the fourth lens, and SAG_{51} is a distance along the optical axis from an intersection of the object-side surface of the fifth lens and the optical axis to a vertex of an effective radius of the object-side surface of the fifth lens. More specifically, SAG_{42} and SAG_{51} may further satisfy: $1.30 < (SAG_{42} + SAG_{51}) / (SAG_{42} - SAG_{51}) < 3.10$. When $1.00 < (SAG_{42} + SAG_{51}) / (SAG_{42} - SAG_{51}) < 3.50$ is satisfied, it is beneficial to avoid the risk of low yield rate due to the large step difference between SAG_{42} and SAG_{51} . At the same time, it is also beneficial to reduce the ghosting phenomenon caused by reflection between the object-side surface of the fifth lens and the image-side surface of the fourth lens.

(38) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $1.00 < (ET_4 + ET_5) / (ET_5 - ET_4) < 4.00$, where ET_4 is an edge thickness of the fourth lens, and ET_5 is an edge thickness of the fifth lens. More specifically, ET_4 and ET_5 may further satisfy: $1.80 < (ET_4 + ET_5) / (ET_5 - ET_4) < 3.50$. When $1.00 < (ET_4 + ET_5) / (ET_5 - ET_4) < 4.00$ is satisfied, the problems, such as difficulty in mass production, due to the thin edge thicknesses of the fourth lens and the fifth lens may be avoided, and the image quality and the relative illumination at the edge field-of-view may be improved by adjusting the edge thicknesses of the fourth and fifth lens.

(39) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $0.50 < CT_5/CT_3 < 2.50$, where CT_5 is a center thickness of the fifth lens along the optical axis, and CT_3 is a center thickness of the third lens along the optical axis. More specifically, CT_5 and CT_3 may further satisfy: $0.90 < CT_5/CT_3 < 2.20$. Satisfying $0.50 < CT_5/CT_3 < 2.50$ may avoid problems, such as difficulty in lens molding, due to the thin center thicknesses of the third lens and the fifth lens along the optical axis, and may also avoid the problem that the overall size of the lens assembly is too long due to the thick center thicknesses of the third lens and the fifth lens along the optical axis.

(40) In an exemplary embodiment, the camera lens group according to the present disclosure may satisfy: $1.00 < f/f_4 < 3.00$, where f is a total effective focal length of the camera lens group, and f_4 is an effective focal length of the fourth lens. More specifically, f and f_4 may further satisfy: $1.30 < f/f_4 < 2.40$. Satisfying $1.00 < f/f_4 < 3.00$ is conducive to controlling the percentage of the effective focal length of the fourth lens in the total effective focal length. In this way, the refractive power of the camera lens group may be more reasonably distributed, and the sensitivity of the

overall camera lens group may be reduced, thereby the processing yield is improved.

(41) In an exemplary embodiment, the camera lens group according to the present disclosure further includes a stop disposed between the object side and the first lens. Optionally, the above camera lens group may further include an optical filter for correcting the color deviation and/or a protective glass for protecting the photosensitive element located on an imaging plane.

(42) The camera lens group according to the above embodiments of the present disclosure may employ a plurality of lenses, such as five lenses as described above. By properly configuring the refractive power of each lens, the surface shape, the center thickness of each lens, and spaced intervals along the optical axis between the lenses, the size of the camera lens group may be effectively reduced, and the workability of the camera lens group may be improved, such that the camera lens group is more advantageous for production processing and may be applied to portable electronic products. The camera lens group configured as described above has the characteristics of ultra-small front end, ultra-large field-of-view, and good image quality, so as to well satisfy the shooting needs of various portable electronic products in different scenarios.

(43) In the embodiments of the present disclosure, at least one of the surfaces of lenses is aspheric, that is, at least one of the object-side surface of the first lens to the image-side surface of the fifth lens is aspheric. The aspheric lens is characterized by a continuous change in curvature from the center of the lens to the periphery of the lens. Unlike a spherical lens having a constant curvature from the center of the lens to the periphery of the lens, the aspheric lens has a better curvature radius characteristic, and has the advantages of improving distortion aberration and improving astigmatic aberration. With aspheric lens, the aberrations that occur during imaging may be eliminated as much as possible, and thus improving the image quality. Optionally, at least one of the object-side surface and the image-side surface of each of the first lens, the second lens, the third lens, the fourth lens and the fifth lens is aspheric. Optionally, the object-side surface and the image-side surface of each of the first lens, the second lens, the third lens, the fourth lens and the fifth lens are aspheric.

(44) However, it will be understood by those skilled in the art that the number of lenses constituting the camera lens group may be varied to achieve the various results and advantages described in this specification without departing from the technical solution claimed by the present disclosure. For example, although the embodiment is described by taking five lenses as an example, the camera lens group is not limited to include five lenses. The camera lens group may also include other numbers of lenses if desired.

(45) Some specific examples of a camera lens group applicable to the above embodiment will be further described below with reference to the accompanying drawings.

Example 1

(46) A camera lens group according to example 1 of the present disclosure is described below with reference to FIG. 1 to FIG. 2D. FIG. 1 shows a schematic structural view of the camera lens group according to example 1 of the present disclosure.

(47) As shown in FIG. 1, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(48) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is concave. The second lens E2 has positive refractive power, an object-side surface S3 thereof is concave, and an image-side surface S4 thereof is convex. The third lens E3 has negative refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and

is finally imaged on the imaging plane S13.

(49) Table 1 is a table illustrating basic parameters of the camera lens group of example 1, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm).

(50) TABLE-US-00001 TABLE 1 Material Surface Surface Radius of Thickness/ Refractive Abbe Focal Conic number type curvature Distance index number length coefficient OBJ Infinite Infinite STO Infinite -0.0260 S1 Aspheric 2.6902 0.3601 1.55 56.1 9.56 -11.8079 S2 Aspheric 5.2867 0.2206 45.8857 S3 Aspheric -10.6685 0.3908 1.55 56.1 4.45 65.0327 S4 Aspheric -2.0031 0.0200 -1.4454 S5 Aspheric 2.3709 0.2798 1.68 19.2 -30.39 -9.4155 S6 Aspheric 2.0248 0.4488 -3.2296 S7 Aspheric -1.8517 0.8285 1.55 56.1 1.44 -3.8951 S8 Aspheric -0.6406 0.0203 -1.7703 S9 Aspheric 10.3343 0.5954 1.62 23.5 -1.37 1.9893 S10 Aspheric 0.7927 0.5008 -6.6153 S11 Infinite 0.2100 1.52 64.2 S12 Infinite 0.4308 S13 Infinite

(51) In this example, a total effective focal length f of the camera lens group is 2.75 mm, a total length TTL of the camera lens group (that is, a distance along the optical axis from the object-side surface S1 of the first lens E1 to the imaging plane S13 of the camera lens group) is 4.31 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 3.07 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 50.8° , and an aperture value F_{no} of the camera lens group is 2.25.

(52) In example 1, the object-side surface and the image-side surface of any one of the first lens E1 to the fifth lens E5 are aspheric. The surface shape x of each aspheric lens may be defined by using, but not limited to, the following aspheric formula:

$$(53) \quad x = \frac{ch^2}{1 + \sqrt{1 - (k+1)c^2h^2}} + \text{.Math. } A_i h^i \quad (1)$$

(54) Where, x is the sag—the axis-component of the displacement of the surface from the aspheric vertex, when the surface is at height h from the optical axis; c is a paraxial curvature of the aspheric surface, $c=1/R$ (that is, the paraxial curvature c is reciprocal of the radius of curvature R in the above Table 1); k is a conic coefficient; A_i is a correction coefficient for the i -th order of the aspheric surface. Table 2 below shows high-order coefficients A4, A6, A8, A10, A12, A14, A16, A18 and A20 applicable to each aspheric surface S1 to S10 in example 1.

(55) TABLE-US-00002 TABLE 2 Surface number A4 A6 A8 A10 A12 S1 1.2116E-02 -6.4255E-01 5.7786E+00 -3.4530E+01 1.1270E+02 S2 -1.5618E-01 -2.3276E-01 -1.7805E-02 3.1810E-01 -5.0225E-01 S3 -1.1861E-01 1.1341E+00 -1.6243E+01 1.1012E+02 -4.0606E+02 S4 -1.0229E-01 -5.3428E-01 6.1981E+00 -3.3985E+01 1.0318E+02 S5 -6.4120E-02 -1.3045E-01 1.0982E+00 -3.7711E+00 7.3820E+00 S6 -3.0425E-02 -1.3183E-01 4.3269E-01 -7.2637E-01 6.8304E-01 S7 -1.6746E-01 1.6379E+00 -8.9337E+00 3.2811E+01 -8.4917E+01 S8 7.3032E-01 -3.1730E+00 1.0425E+01 -2.6322E+01 4.8337E+01 S9 1.9758E-01 -5.4287E-01 5.2132E-01 -4.8052E-02 -6.5167E-01 S10 1.3833E-02 -7.0650E-02 3.6308E-02 1.9569E-02 -4.2367E-02 Surface number A14 A16 A18 A20 S1 -1.8945E+02 1.2800E+02 0.0000E+00 0.0000E+00 S2 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 S3 5.5610E+02 1.6792E+03 -9.8591E+03 2.2140E+04 S4 -1.4720E+02 -8.6863E+01 7.9140E+02 -1.5418E+03 S5 -9.0079E+00 6.8472E+00 -2.9915E+00 5.7220E-01 S6 -3.3925E-01 6.5987E-02 5.9160E-03 -2.7754E-03 S7 1.5775E+02 -2.1063E+02 2.0140E+02 -1.3640E+02 S8 -6.3701E+01 6.0275E+01 -4.0849E+01 1.9609E+01 S9 1.0974E+00 -1.0257E+00 6.3659E-01 -2.7453E-01 S10 3.2337E-02 -1.5156E-02 4.8471E-03 -1.0937E-03

(56) FIG. 2A illustrates the longitudinal aberration curves of the camera lens group according to example 1, representing the deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 2B illustrates the astigmatic curves of the camera lens group according to example 1, representing the curvatures of a tangential plane and the curvatures

of a sagittal plane. FIG. 2C illustrates a distortion curve of the camera lens group according to example 1, representing the amounts of distortion corresponding to different field-of-views. FIG. 2D illustrates a lateral color curve of the camera lens group according to example 1, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 2A to FIG. 2D that the camera lens group provided in example 1 may achieve good image quality.

Example 2

(57) A camera lens group according to example 2 of the present disclosure is described below with reference to FIG. 3 to FIG. 4D. In this example and the following examples, for the purpose of brevity, the description of parts similar to those in example 1 will be omitted. FIG. 3 shows a schematic structural view of the camera lens group according to example 2 of the present disclosure.

(58) As shown in FIG. 3, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(59) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is concave. The second lens E2 has negative refractive power, an object-side surface S3 thereof is convex, and an image-side surface S4 thereof is concave. The third lens E3 has positive refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(60) In this example, a total effective focal length f of the camera lens group is 3.11 mm, a total length TTL of the camera lens group is 4.37 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 3.27 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 46.1° , and an aperture value F_{no} of the camera lens group is 2.24.

(61) Table 3 is a table illustrating basic parameters of the camera lens group of example 2, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 4 shows high-order coefficients applicable to each aspheric surface in example 2, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(62) TABLE-US-00003

TABLE 3	Material	Surface	Surface Radius of Curvature	Thickness/ Distance	Refractive index	Abbe number	Focal length	Conic coefficient	OBJ	Infinite	Infinite
STO	Infinite	-0.1018	S1 Aspheric	1.8219	0.5700	1.55	56.1	4.85	-3.6259	S2 Aspheric	5.1939
		0.2400	-88.6691	S3 Aspheric	9.9153	0.2200	1.67	20.4	-6.47	-0.6749	S4 Aspheric
		-35.7741	S5 Aspheric	2.2218	0.3793	1.55	56.1	6.92	-1.1801	S6 Aspheric	5.0646
		-21.9222	S7 Aspheric	-7.2275	0.7629	1.55	56.1	1.38	11.6728	S8 Aspheric	-0.7098
		-5.2796	S9 Aspheric	11.4222	0.4000	1.54	55.9	-1.28	16.6558	S10 Aspheric	0.6413
		-4.8346	S11	Infinite	0.2100	1.52	64.2	S12	Infinite	0.4914	S13
											Infinite

(63) TABLE-US-00004

TABLE 4	Surface number	A4	A6	A8	A10	A12	S1	3.1695E-02
	S2	5.2247E-01	-6.4515E+00	4.1199E+01	-1.5695E+02	S2	5.5258E-03	-7.8543E-01
	S3	5.8928E+00	-3.5996E+01	1.3660E+02	S3	-1.5107E-01	4.8949E-02	-1.8439E+00
	S4	9.0226E+00	-2.9302E+01	S4	-1.7599E-01	2.9993E-01	1.0546E+00	-9.2579E+00
	S5	2.8363E+01	S5	-4.7627E-01	9.3770E-01	-5.9945E-01	-2.5891E+00	9.0540E+00
	S6	-1.3589E-01	-6.2987E-02	4.9425E-01	-1.0160E+00	1.0833E+00	S7	3.0001E-02
	S8	-2.6374E-01	3.3757E-01	-1.0978E-01	-6.0562E-01	S8	-3.9265E-01	1.0746E+00

-2.5939E+00 4.0484E+00 -4.0400E+00 S9 -1.1952E-01 -1.2062E-01 1.2062E-01
1.0870E-01 -2.5076E-01 S10 -1.1157E-01 4.6594E-02 -1.1441E-02 5.4644E-03
-6.6694E-03 Surface number A14 A16 A18 A20 S1 3.6030E+02 -4.8344E+02 3.4185E+02
-9.4647E+01 S2 -3.2507E+02 4.7102E+02 -3.7989E+02 1.3104E+02 S3 5.9343E+01
-7.1639E+01 4.7712E+01 -1.3188E+01 S4 -4.8606E+01 4.9473E+01 -2.8017E+01
6.8138E+00 S5 -1.4435E+01 1.3069E+01 -6.4565E+00 1.3455E+00 S6 -6.4105E-01
1.8101E-01 -1.4069E-02 1.4288E-03 S7 1.2836E+00 -1.1774E+00 5.2418E-01 -9.1189E-02
S8 2.5615E+00 -9.8709E-01 2.0955E-01 -1.8758E-02 S9 2.0397E-01 -9.6226E-02
2.8981E-02 -5.6653E-03 S10 4.7905E-03 -2.0012E-03 5.2295E-04 -8.7114E-05

(64) FIG. 4A illustrates the longitudinal aberration curves of the camera lens group according to example 2, representing the deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 4B illustrates the astigmatic curves of the camera lens group according to example 2, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 4C illustrates a distortion curve of the camera lens group according to example 2, representing the amounts of distortion corresponding to different field-of-views. FIG. 4D illustrates a lateral color curve of the camera lens group according to example 2, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 4A to FIG. 4D that the camera lens group provided in example 2 may achieve good image quality.

Example 3

(65) A camera lens group according to example 3 of the present disclosure is described below with reference to FIG. 5 to FIG. 6D. FIG. 5 shows a schematic structural view of the camera lens group according to example 3 of the present disclosure.

(66) As shown in FIG. 5, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(67) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is concave. The second lens E2 has positive refractive power, an object-side surface S3 thereof is concave, and an image-side surface S4 thereof is convex. The third lens E3 has negative refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(68) In this example, a total effective focal length f of the camera lens group is 2.67 mm, a total length TTL of the camera lens group is 3.96 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 2.57 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 46.0° , and an aperture value F_{no} of the camera lens group is 2.24.

(69) Table 5 is a table illustrating basic parameters of the camera lens group of example 3, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 6 shows high-order coefficients applicable to each aspheric surface in example 3, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(70) TABLE-US-00005 TABLE 5 Material Surface Surface Radius of Thickness/ Refractive Abbe Focal Conic number type curvature Distance index number length coefficient OBJ Infinite Infinite STO Infinite -0.0260 S1 Aspheric 2.2484 0.3500 1.55 56.1 7.87 -10.9991 S2 Aspheric 4.4583 0.1710 41.5002 S3 Aspheric -17.9237 0.3413 1.67 20.4 3.27 -99.0000 S4 Aspheric -1.6347

0.0200 -3.3048 S5 Aspheric 2.3290 0.2689 1.55 56.1 -6.02 -18.1918 S6 Aspheric 1.4134 0.4109
-3.6000 S7 Aspheric -2.2396 0.7251 1.55 56.1 1.46 -3.3706 S8 Aspheric -0.6774 0.0693 -1.6267
S9 Aspheric 199.3766 0.5716 1.54 55.9 -1.37 16.6555 S10 Aspheric 0.8792 0.4510 -1.0000 S11
Infinite 0.2100 1.52 64.2 S12 Infinite 0.3753 S13 Infinite

(71) TABLE-US-00006 TABLE 6 Surface number A4 A6 A8 A10 A12 S1 -6.9589E-03
-3.4413E-01 1.4064E+00 -1.1206E+01 4.8668E+01 S2 -3.0247E-01 -3.8346E-01
-5.4342E-01 1.7439E+00 -1.7119E+00 S3 -1.9953E-01 2.0792E+00 -5.8286E+01
8.1819E+02 -7.5078E+03 S4 -5.9535E-02 -1.8148E+00 3.2954E+01 -3.6940E+02
2.6729E+03 S5 -1.2105E-01 1.8286E-01 -1.0813E+00 3.7254E+00 -7.1000E+00 S6
-1.6302E-01 2.3880E-01 -3.7584E-01 4.0855E-01 9.0457E-03 S7 5.1916E-02
6.6778E-01 -8.7114E+00 5.6983E+01 -2.3491E+02 S8 1.1196E+00 -6.2327E+00 2.1570E+01
-4.8947E+00I.sup. 6.7853E+01 S9 8.3029E-01 -5.9918E+00 2.2791E+01 -5.8560E+01
1.0645E+02 S10 -8.1293E-01 1.2279E+00 -1.6653E+00 1.8030E+00 -1.5250E+00 Surface
number A14 A16 A18 A20 S1 -1.0961E+02 9.9233E+01 0.0000E+00 0.0000E+00 S2
0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 S3 4.6720E+04 -2.0003E+05 5.8907E+05
-1.1707E+06 S4 -1.3125E+04 4.4760E+04 -1.0614E+05 1.7167E+05 S5 7.1123E+00
-2.8950E+00 -1.2429E+00 1.4022E+00 S6 -7.2085E-01 8.9040E-01 -4.4770E-01
8.4144E-02 S7 6.5771E+02 -1.2903E+03 1.7945E+03 -1.7620E+03 S8 -3.7894E+01
-5.0850E+01 1.3947E+02 -1.5661E+02 S9 -1.4004E+02 1.3502E+02 -9.5870E+01
4.9951E+01 S10 1.0101E+00 -5.2573E-01 2.1361E-01 -6.6537E-02

(72) FIG. 6A illustrates the longitudinal aberration curves of the camera lens group according to example 3, representing the deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 6B illustrates the astigmatic curves of the camera lens group according to example 3, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 6C illustrates a distortion curve of the camera lens group according to example 3, representing the amounts of distortion corresponding to different field-of-views. FIG. 6D illustrates a lateral color curve of the camera lens group according to example 3, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 6A to FIG. 6D that the camera lens group provided in example 3 may achieve good image quality.

Example 4

(73) A camera lens group according to example 4 of the present disclosure is described below with reference to FIG. 7 to FIG. 8D. FIG. 7 shows a schematic structural view of the camera lens group according to example 4 of the present disclosure.

(74) As shown in FIG. 7, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(75) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is concave. The second lens E2 has positive refractive power, an object-side surface S3 thereof is concave, and an image-side surface S4 thereof is convex. The third lens E3 has negative refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(76) In this example, a total effective focal length f of the camera lens group is 2.46 mm, a total length TTL of the camera lens group is 3.81 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 2.73 mm, half of a maximum field-

of-view-Semi-FOV of the camera lens group is 51.1°, and an aperture value Fno of the camera lens group is 2.24.

(77) Table 7 is a table illustrating basic parameters of the camera lens group of example 4, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 8 shows high-order coefficients applicable to each aspheric surface in example 4, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(78) TABLE-US-00007 TABLE 7 Material Surface Surface Radius of Thickness/ Refractive Abbe Focal Conic number type curvature Distance index number length coefficient OBJ Infinite Infinite STO Infinite -0.0260 S1 Aspheric 2.1974 0.3500 1.55 56.1 7.89 -12.0654 S2 Aspheric 4.2331 0.1211 42.0591 S3 Aspheric -6.5875 0.3280 1.67 20.4 2.78 -98.8357 S4 Aspheric -1.2555 0.0200 -5.0689 S5 Aspheric 2.5462 0.2736 1.55 56.1 -4.75 -18.0552 S6 Aspheric 1.3594 0.3727 -3.7048 S7 Aspheric -2.3347 0.7874 1.55 56.1 1.48 -2.2314 S8 Aspheric -0.6954 0.0200 -1.4169 S9 Aspheric 3.7394 0.4991 1.54 55.9 -1.53 0.0078 S10 Aspheric 0.7378 0.4561 -1.0000 S11 Infinite 0.2100 1.52 64.2 S12 Infinite 0.3718 S13 Infinite

(79) TABLE-US-00008 TABLE 8 Surface number A4 A6 A8 A10 A12 S1 -6.3432E-03 -4.3213E-01 1.9655E+00 -1.8057E+01 9.0340E+01 S2 -3.5096E-01 -7.7642E-01 7.1380E-01 -8.7087E-01 4.2697E-01 S3 -2.9796E-01 3.8303E+00 -1.1178E+02 1.7257E+03 -1.7266E+04 S4 -1.2540E-01 -2.1258E+00 4.4840E+01 -5.4961E+02 4.3554E+03 S5 -1.7291E-01 4.6396E-01 -3.3787E+00 1.8338E+01 -5.7242E+01 S6 -2.8578E-01 8.2478E-01 -2.6377E+00 6.8833E+00 -1.2214E+01 S7 -8.7049E-03 1.1536E+00 -1.2630E+01 8.3372E+01 -3.5584E+02 S8 7.1945E-01 -4.4151E+00 1.9597E+01 -6.7995E+01 1.8530E+02 S9 1.0348E-01 -3.2262E+00 1.4589E+01 -3.9901E+01 7.4239E+01 S10 -1.2322E+00 2.3930E+00 -4.0264E+00 5.2346E+00 -5.1381E+00 Surface number A14 A16 A18 A20 S1 -2.3713E+02 2.5152E+02 0.0000E+00 0.0000E+00 S2 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 S3 1.1776E+05 -5.5587E+05 1.8152E+06 -4.0241E+06 S4 -2.2982E+04 8.3001E+04 -2.0658E+05 3.4875E+05 S5 1.0531E+02 -1.1629E+02 7.1538E+01 -1.8858E+01 S6 1.3851E+01 -9.7340E+00 3.9240E+00 -7.0341E-01 S7 1.0337E+03 -2.1030E+03 3.0327E+03 -3.0896E+03 S8 -3.8856E+02 6.0964E+02 -6.9823E+02 5.6994E+02 S9 -9.8481E+01 9.5360E+01 -6.8044E+01 3.5716E+01 S10 3.7914E+00 -2.1000E+00 8.6983E-01 -2.6702E-01

(80) FIG. 8A illustrates the longitudinal aberration curves of the camera lens group according to example 4, representing the deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 8B illustrates the astigmatic curves of the camera lens group according to example 4, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 8C illustrates a distortion curve of the camera lens group according to example 4, representing the amounts of distortion corresponding to different field-of-views. FIG. 8D illustrates a lateral color curve of the camera lens group according to example 4, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 8A to FIG. 8D that the camera lens group provided in example 4 may achieve good image quality.

Example 5

(81) A camera lens group according to example 5 of the present disclosure is described below with reference to FIG. 9 to FIG. 10D. FIG. 9 shows a schematic structural view of the camera lens group according to example 5 of the present disclosure.

(82) As shown in FIG. 9, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(83) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is concave. The second lens E2 has negative refractive power, an

object-side surface S3 thereof is convex, and an image-side surface S4 thereof is concave. The third lens E3 has positive refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(84) In this example, a total effective focal length f of the camera lens group is 3.12 mm, a total length TTL of the camera lens group is 4.39 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 3.00 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 43.2° , and an aperture value F_{no} of the camera lens group is 2.33.

(85) Table 9 is a table illustrating basic parameters of the camera lens group of example 5, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 10 shows high-order coefficients applicable to each aspheric surface in example 5, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(86) TABLE-US-00009 TABLE 9 Material Surface Surface Radius of Thickness/ Refractive Abbe Focal Conic number type curvature Distance index number length coefficient OBJ Infinite Infinite STO Infinite -0.1118 S1 Aspheric 1.6382 0.5429 1.55 56.1 4.54 -2.0110 S2 Aspheric 4.2734 0.2729 -24.3347 S3 Aspheric 22.9364 0.2277 1.67 20.4 -5.84 53.1732 S4 Aspheric 3.3102 0.0504 -53.1243 S5 Aspheric 2.7651 0.4190 1.55 56.1 7.63 -0.2579 S6 Aspheric 7.7783 0.3761 -80.3934 S7 Aspheric -10.7658 0.7902 1.55 56.1 1.38 30.8484 S8 Aspheric -0.7224 0.0996 -4.9497 S9 Aspheric 11.1402 0.4007 1.54 55.9 -1.30 17.2695 S10 Aspheric 0.6480 0.5051 -4.7244 S11 Infinite 0.2100 1.52 64.2 S12 Infinite 0.4944 S13 Infinite

(87) TABLE-US-00010 TABLE 10 Surface number A4 A6 A8 A10 A12 S1 5.8210E-02 -1.2133E-01 1.8615E+00 -1.6409E+01 8.2815E+01 S2 -2.8978E-03 -3.7810E-01 2.6847E+00 -1.6616E+01 6.1740E+01 S3 -2.1669E-01 6.6102E-01 -5.8092E+00 2.6030E+01 -7.7379E+01 S4 -1.9807E-01 4.9007E-01 -3.3140E-01 -3.3607E+00 1.3101E+01 S5 -4.6177E-01 9.7275E-01 -1.1317E+00 9.0917E-03 2.2830E+00 S6 -1.4094E-01 -5.1471E-02 4.0703E-01 -7.0451E-01 5.9805E-01 S7 -2.0486E-02 -8.3363E-02 -1.8872E-01 6.7468E-01 -9.9431E-01 S8 -2.9470E-01 6.1506E-01 -1.3361E+00 1.8737E+00 -1.6596E+00 S9 -1.0645E-01 -2.1300E-01 3.4154E-01 -1.8433E-01 -9.3797E-04 S10 -1.0963E-01 2.8183E-02 2.5745E-02 -3.1543E-02 1.5572E-02 Surface number A14 A16 A18 A20 S1 -2.4954E+02 4.4218E+02 -4.2546E+02 1.7156E+02 S2 -1.4399E+02 2.0429E+02 -1.6159E+02 5.4989E+01 S3 1.4874E+02 -1.7935E+02 1.2404E+02 -3.7011E+01 S4 -2.3571E+01 2.3725E+01 -1.2794E+01 2.8942E+00 S5 -3.8650E+00 3.1857E+00 -1.3714E+00 2.4728E-01 S6 -2.7593E-01 9.4117E-02 -5.0913E-02 1.7741E-02 S7 9.3449E-01 -5.9237E-01 2.2262E-01 -3.5727E-02 S8 9.3945E-01 -3.2547E-01 6.2288E-02 -5.0271E-03 S9 5.8784E-02 -3.7555E-02 1.2537E-02 -2.5383E-03 S10 -3.9964E-03 3.7302E-04 7.8562E-05 -3.0105E-05

(88) FIG. 10A illustrates the longitudinal aberration curves of the camera lens group according to example 5, representing the deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 10B illustrates the astigmatic curves of the camera lens group according to example 5, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 10C illustrates a distortion curve of the camera lens group according to example 5, representing the amounts of distortion corresponding to different field-of-views. FIG. 10D illustrates a lateral color curve of the camera lens group according to example 5,

representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 10A to FIG. 10D that the camera lens group provided in example 5 may achieve good image quality.

Example 6

(89) A camera lens group according to example 6 of the present disclosure is described below with reference to FIG. 11 to FIG. 12D. FIG. 11 shows a schematic structural view of the camera lens group according to example 6 of the present disclosure.

(90) As shown in FIG. 11, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(91) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is convex. The second lens E2 has positive refractive power, an object-side surface S3 thereof is convex, and an image-side surface S4 thereof is convex. The third lens E3 has negative refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(92) In this example, a total effective focal length f of the camera lens group is 2.01 mm, a total length TTL of the camera lens group is 3.52 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 2.45 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 55.6° , and an aperture value F_{no} of the camera lens group is 2.29.

(93) Table 11 is a table illustrating basic parameters of the camera lens group of example 6, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 12 shows high-order coefficients applicable to each aspheric surface in example 6, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(94) TABLE-US-00011

TABLE	11	Material	Surface	Surface Radius of Curvature	Thickness/Distance	Refractive Index	Abbe Number
STO	Infinite	0.0140	S1	Aspheric	7.4724	0.3700	1.55 56.1 10.06 -93.1121
			S2	Aspheric	-20.3821		
	0.0929	15.2584	S3	Aspheric	5.3472	0.2200	1.55 56.1 4.49 -88.8751
			S4	Aspheric	-4.4514	0.0200	
	18.4831		S5	Aspheric	1.4054	0.2200	1.68 19.2 -10.23 -4.1750
			S6	Aspheric	1.0945	0.2796	-7.0648
			S7	Aspheric	-1.9610	0.8700	1.55 56.1 1.26 2.1991
			S8	Aspheric	-0.5874	0.0200	-2.2799
			S9	Aspheric	1.1700	0.3362	1.67 20.4 -1.74 -0.9812
			S10	Aspheric	0.5146	0.5621	-3.9158
			S11	Infinite			
	0.2100	1.52	64.2	S12	Infinite	0.3201	S13 Infinite

(95) TABLE-US-00012

TABLE	12	Surface number	A4	A6	A8	A10	A12	S1
			-4.4901E-01	-8.5335E-01	0.0000E+00	0.0000E+00	S2	-6.0827E-01 -6.1879E+00
		3.6238E+01	-1.3444E+02	2.9808E+02	S3	7.7330E-01	-9.2915E+00	3.4858E+01
		-1.7485E+02	8.8433E+02	S4	1.2842E+00	-4.8828E+00	2.5233E+00	4.5380E+01
		-2.2841E+02	S5	-5.9668E-01	3.7938E-01	-1.1803E+00	1.8597E+01	-7.4162E+01
		-3.7506E-01	8.7600E-01	-3.1807E+00	1.0209E+01	-1.7050E+01	S7	3.2754E-01
		1.1116E-01	-2.1480E+00	8.6864E+00	-1.9051E+01	S8	2.1394E-01	-2.1122E+00 7.6183E+00
		-1.8004E+01	2.8520E+01	S9	-4.9607E-01	4.7131E-01	-5.6712E-01	5.5573E-01
		-3.7986E-01	S10	-1.2791E-01	3.8482E-02	5.5971E-03	-1.9139E-02	1.4684E-02
		Surface number	A14	A16	A18	A20	S1	0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00
			-2.6757E+02	0.0000E+00	0.0000E+00	0.0000E+00	S3	-3.1750E+03 7.4030E+03
			-9.6689E+03	5.2245E+03	S4	5.7289E+02	-8.9045E+02	8.5247E+02 -3.7334E+02 S5

1.2739E+02 -1.2178E+02 8.3177E+01 -3.3925E+01 S6 6.9388E+00 1.6477E+01
-2.2332E+01 8.3082E+00 S7 2.6371E+01 -2.3194E+01 1.1872E+01 -2.6905E+00 S8
-2.9527E+01 1.9162E+01 -7.0252E+00 1.1009E+00 S9 1.7270E-01 -5.0185E-02
8.5256E-03 -6.4564E-04 S10 -6.5545E-03 1.7622E-03 -2.6277E-04 1.6652E-05

(96) FIG. 12A illustrates the longitudinal aberration curves of the camera lens group according to example 6, representing the deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 12B illustrates the astigmatic curves of the camera lens group according to example 6, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 12C illustrates a distortion curve of the camera lens group according to example 6, representing the amounts of distortion corresponding to different field-of-views. FIG. 12D illustrates a lateral color curve of the camera lens group according to example 6, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 12A to FIG. 12D that the camera lens group provided in example 6 may achieve good image quality.

Example 7

(97) A camera lens group according to example 7 of the present disclosure is described below with reference to FIG. 13 to FIG. 14D. FIG. 13 shows a schematic structural view of the camera lens group according to example 7 of the present disclosure.

(98) As shown in FIG. 13, the camera lens group includes a stop ST0, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(99) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is concave. The second lens E2 has positive refractive power, an object-side surface S3 thereof is convex, and an image-side surface S4 thereof is convex. The third lens E3 has negative refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(100) In this example, a total effective focal length f of the camera lens group is 2.02 mm, a total length TTL of the camera lens group is 3.47 mm, half of a diagonal length $ImgH$ of an effective pixel area on the imaging plane **S13** of the camera lens group is 2.30 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 52.5°, and an aperture value Fno of the camera lens group is 2.28.

(101) Table 13 is a table illustrating basic parameters of the camera lens group of example 7, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 14 shows high-order coefficients applicable to each aspheric surface in example 7, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(102) TABLE-US-00013 TABLE 13 Material Surface Surface Radius of Thickness/ Refractive Abbe Focal Conic number type curvature Distance index number length coefficient OBJ Infinite Infinite STO Infinite 0.1800 S1 Aspheric 6.2242 0.3453 1.55 56.1 35.78 -39.9156 S2 Aspheric 8.9564 0.0260 93.9210 S3 Aspheric 18.5694 0.2763 1.55 56.1 3.89 -99.0022 S4 Aspheric -2.3866 0.0250 3.6238 S5 Aspheric 1.2140 0.2100 1.68 19.2 -58.01 -1.3973 S6 Aspheric 1.0954 0.2718 -3.9401 S7 Aspheric -1.9105 0.8500 1.55 56.1 1.49 1.5974 S8 Aspheric -0.6607 0.0220 -2.7504 S9 Aspheric 1.0956 0.3463 1.68 19.2 -2.02 -0.5753 S10 Aspheric 0.5308 0.5623 -3.8590 S11 Infinite 0.2100 1.52 64.2 S12 Infinite 0.3282 S13 Infinite

(103) TABLE-US-00014 TABLE 14 Surface number A4 A6 A8 A10 A12 S1 -5.8978E-01

7.0649E+00 -1.1905E+02 1.1974E+03 -7.7432E+03 S2 7.0895E-02 -4.4775E+00
-9.1856E+01 9.4416E+02 -4.1818E+03 S3 1.3333E+00 -6.1182E+00 -6.6335E+01
5.7812E+02 -1.9516E+03 S4 9.1592E-01 2.2162E+00 -3.2969E+01 1.4496E+02 -3.6874E+02
S5 -3.0573E-01 1.9898E-02 -5.6401E+00 3.4206E+01 -8.9544E+01 S6 -1.1511E-01
3.6840E-01 -5.1326E+00 2.1454E+01 -4.5541E+01 S7 4.0875E-01 5.4593E-01 -1.9558E+00
7.1719E-01 5.3320E+00 S8 -2.0127E-01 3.8874E-01 -9.3632E-01 1.8899E+00 -2.0990E+00
S9 -7.4893E-01 1.4759E+00 -3.4879E+00 5.7580E+00 -6.2984E+00 S10 -7.7496E-02
-9.9233E-02 2.2714E-01 -2.4454E-01 1.5683E-01 Surface number A14 A16 A18 A20 S1
3.1914E+04 -7.9937E+04 1.0997E+05 -6.3524E+04 S2 1.0890E+04 -1.7168E+04 1.5014E+04
-5.5431E+03 S3 3.6218E+03 -3.9124E+03 2.3188E+03 -5.8476E+02 S4 5.8220E+02
-5.5716E+02 2.9436E+02 -6.5528E+01 S5 1.2482E+02 -9.6622E+01 3.9434E+01 -6.6662E+00
S6 5.5767E+01 -4.0103E+01 1.5844E+01 -2.6704E+00 S7 -1.0729E+01 9.2828E+00
-3.9431E+00 6.7156E-01 S8 8.9824E-01 3.4943E-01 -4.5410E-01 1.1534E-01 S9
4.4165E+00 -1.8961E+00 4.5111E-01 -4.5324E-02 S10 -6.1838E-02 1.4651E-02
-1.9162E-03 1.0652E-04

(104) FIG. 14A illustrates the longitudinal aberration curves of the camera lens group according to example 7, representing deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 14B illustrates the astigmatic curves of the camera lens group according to example 7, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 14C illustrates a distortion curve of the camera lens group according to example 7, representing the amounts of distortion corresponding to different field-of-views. FIG. 14D illustrates a lateral color curve of the camera lens group according to example 7, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 14A to FIG. 14D that the camera lens group provided in example 7 may achieve good image quality.

Example 8

(105) A camera lens group according to example 8 of the present disclosure is described below with reference to FIG. 15 to FIG. 16D. FIG. 15 shows a schematic structural view of the camera lens group according to example 8 of the present disclosure.

(106) As shown in FIG. 15, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(107) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is convex. The second lens E2 has positive refractive power, an object-side surface S3 thereof is concave, and an image-side surface S4 thereof is convex. The third lens E3 has negative refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(108) In this example, a total effective focal length f of the camera lens group is 2.02 mm, a total length TTL of the camera lens group is 3.26 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 2.40 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 55.0° , and an aperture value F_{no} of the camera lens group is 2.50.

(109) Table 15 is a table illustrating basic parameters of the camera lens group of example 8, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 16 shows high-order coefficients applicable to each aspheric surface in

example 8, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(110) TABLE-US-00015 TABLE 15 Material Surface Surface Radius of Thickness/ Refractive Abbe Focal Conic number type curvature Distance index number length coefficient OBJ Infinite Infinite STO Infinite -0.0117 S1 Aspheric 2.4343 0.2809 1.55 56.1 4.45 -10.8985 S2 Aspheric -1553.3333 0.1149 90.0000 S3 Aspheric -2.3967 0.2589 1.55 56.1 6.85 -24.4495 S4 Aspheric -1.5166 0.0300 -6.7595 S5 Aspheric 1.7930 0.2150 1.68 19.2 -42.21 -10.3924 S6 Aspheric 1.6055 0.2728 -2.1154 S7 Aspheric -1.2870 0.6689 1.55 56.1 1.37 -10.1716 S8 Aspheric -0.5610 0.0300 -1.6027 S9 Aspheric 1.1733 0.3280 1.62 23.5 -1.62 -0.9053 S10 Aspheric 0.4913 0.4517 -4.2948 S11 Infinite 0.2100 1.52 64.2 S12 Infinite 0.3989 S13 Infinite

(111) TABLE-US-00016 TABLE 16 Surface number A4 A6 A8 A10 A12 S1 -2.5286E-01 3.1170E+00 -1.1051E+02 2.0221E+03 -2.3750E+04 S2 -6.1017E-01 -1.1571E+00 -8.1833E+00 3.7109E+02 -5.4323E+03 S3 -5.0149E-01 -5.2419E-01 7.1213E+00 1.1648E+01 -6.8504E+02 S4 2.7536E-01 -1.1674E+01 1.3102E+02 -8.2080E+02 3.3741E+03 S5 3.4014E-01 -7.8168E+00 6.0865E+01 -3.1076E+02 1.0772E+03 S6 1.5760E-01 -2.3519E+00 8.6479E+00 -1.9254E+01 2.6270E+01 S7 -1.6758E-01 1.3433E+00 -8.0516E+00 4.7254E+01 -1.8796E+02 S8 7.3165E-01 -3.8818E+00 1.2496E+01 -3.0344E+01 5.8647E+01 S9 -6.7912E-01 1.3025E+00 -4.0068E+00 9.7517E+00 -1.6604E+01 S10 -5.7499E-02 -3.0428E-01 8.0661E-01 -1.2485E+00 1.3715E+00 Surface number A14 A16 A18 A20 S1 1.7903E+05 -8.3729E+05 2.2039E+06 -2.4782E+06 S2 4.2784E+04 -1.8938E+05 4.4176E+05 -4.1722E+05 S3 7.2802E+03 -3.7405E+04 9.3082E+04 -8.8528E+04 S4 -9.0967E+03 1.5224E+04 -1.4395E+04 6.0047E+03 S5 -2.5089E+03 3.7531E+03 -3.2544E+03 1.2360E+03 S6 -2.0085E+01 6.5059E+00 -2.3320E-02 -5.3823E-02 S7 5.2316E+02 -1.0352E+03 1.4082E+03 -1.2346E+03 S8 -9.2910E+01 1.2186E+02 -1.2177E+02 8.1604E+01 S9 1.9854E+01 -1.6839E+01 1.0150E+01 -4.3096E+00 S10 -1.1178E+00 6.7980E-01 -3.0537E-01 9.9228E-02

(112) FIG. 16A illustrates the longitudinal aberration curves of the camera lens group according to example 8, representing deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 16B illustrates the astigmatic curves of the camera lens group according to example 8, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 16C illustrates a distortion curve of the camera lens group according to example 8, representing the amounts of distortion corresponding to different field-of-views. FIG. 16D illustrates a lateral color curve of the camera lens group according to example 8, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 16A to FIG. 16D that the camera lens group provided in example 8 may achieve good image quality.

Example 9

(113) A camera lens group according to example 9 of the present disclosure is described below with reference to FIG. 17 to FIG. 18D. FIG. 17 shows a schematic structural view of the camera lens group according to example 9 of the present disclosure.

(114) As shown in FIG. 17, the camera lens group includes a stop STO, a first lens E1, a second lens E2, a third lens E3, a fourth lens E4, a fifth lens E5, an optical filter E6 and an imaging plane S13, which are sequentially arranged from an object side to an image side.

(115) The first lens E1 has positive refractive power, an object-side surface S1 thereof is convex, and an image-side surface S2 thereof is concave. The second lens E2 has negative refractive power, an object-side surface S3 thereof is concave, and an image-side surface S4 thereof is concave. The third lens E3 has positive refractive power, an object-side surface S5 thereof is convex, and an image-side surface S6 thereof is concave. The fourth lens E4 has positive refractive power, an object-side surface S7 thereof is concave, and an image-side surface S8 thereof is convex. The fifth

lens E5 has negative refractive power, an object-side surface S9 thereof is convex, and an image-side surface S10 thereof is concave. The optical filter E6 has an object-side surface S11 and an image-side surface S12. Light from an object sequentially passes through the respective surfaces S1 to S12 and is finally imaged on the imaging plane S13.

(116) In this example, a total effective focal length f of the camera lens group is 3.19 mm, a total length TTL of the camera lens group is 4.40 mm, half of a diagonal length ImgH of an effective pixel area on the imaging plane S13 of the camera lens group is 3.14 mm, half of a maximum field-of-view Semi-FOV of the camera lens group is 43.8° , and an aperture value F_{no} of the camera lens group is 2.24.

(117) Table 17 is a table illustrating basic parameters of the camera lens group of example 9, wherein the units for the radius of curvature, the thickness/distance and the focal length are millimeter (mm). Table 18 shows high-order coefficients applicable to each aspheric surface in example 9, wherein the surface shape of each aspheric surface may be defined by the formula (1) given in the above example 1.

(118) TABLE-US-00017

TABLE 17	Material	Surface	Surface Radius of Curvature	Thickness/ Distance	Refractive index	Focal length	Conic coefficient	OBJ	Infinite
Infinite	STO	Infinite	-0.1094	S1	Aspheric	1.6826	0.5460	1.55	56.1
4.25	-2.5667	S2	Aspheric	5.4064	0.2698	-15.9416	S3	Aspheric	-13.6268
0.2249	1.67	20.4	-5.63	99.0000	S4	Aspheric	5.2019	0.1000	-63.1108
S5	Aspheric	3.6991	0.3256	1.55	56.1	7.04	-0.6643	S6	Aspheric
93.9505	0.5036	-99.0000	S7	Aspheric	-4.5769	0.6757	1.55	56.1	1.36
2.7092	S8	Aspheric	-0.6713	0.1000	-4.3735	S9	Aspheric	10.7535	0.3870
1.54	55.9	-1.24	12.4168	S10	Aspheric	0.6205	0.5193	-5.0126	S11
Infinite	0.2276	1.52	64.2	S12	Infinite	0.5206	S13	Infinite	

(119) TABLE-US-00018

TABLE 18	Surface number	A4	A6	A8	A10	A12	S1
6.8084E-02	7.1022E-02	-9.4396E-01	6.7687E+00	-3.1966E+01	S2	-3.3759E-02	7.7038E-03
-9.9214E-01	6.1301E+00	-2.5509E+01	S3	-2.0093E-01	2.1389E-01	-5.5945E-01	-4.4035E+00
3.0133E+01	S4	-2.3938E-01	7.8670E-01	-2.1157E+00	3.2590E+00	-2.1523E+00	S5
-3.3306E-01	7.7140E-01	-1.7405E+00	3.5792E+00	-6.0506E+00	S6	-1.6684E-01	1.0001E-01
-1.4939E-01	4.2064E-01	-1.0113E+00	S7	-4.7382E-02	-1.0232E-01	1.8203E-01	-2.4939E-01
4.4051E-02	S8	-3.0917E-01	6.0939E-01	-1.2672E+00	1.8948E+00	-1.9447E+00	S9
-1.2445E-01	9.0584E-03	4.3006E-02	-3.3466E-02	1.3140E-02	S10	-9.5126E-02	4.9955E-02
-2.0693E-02	6.0659E-03	-1.2690E-03	Surface number	A14	A16	A18	A20
S1	9.8158E+01	-1.8809E+02	2.0262E+02	-9.3404E+01	S2	6.3793E+01	-9.5216E+01
7.6771E+01	-2.5680E+01	S3	-9.2043E+01	1.5524E+02	-1.4162E+02	5.4495E+01	S4
-1.7367E+00	4.7930E+00	-3.8038E+00	1.1035E+00	S5	7.3737E+00	-5.8742E+00	2.7262E+00
-5.6030E-01	S6	1.5348E+00	-1.4263E+00	7.4535E-01	-1.6197E-01	S7	3.2655E-01
-4.1048E-01	2.0207E-01	-3.6626E-02	S8	1.3064E+00	-5.3185E-01	1.1771E-01	-1.0852E-02
S9	-3.0356E-03	4.1396E-04	-3.0815E-05	9.6487E-07	S10	1.8356E-04	-1.7554E-05
1.0277E-06	-2.8615E-08						

(120) FIG. 18A illustrates the longitudinal aberration curves of the camera lens group according to example 9, representing deviations of focal points converged by light of different wavelengths after passing through the lens assembly. FIG. 18B illustrates the astigmatic curves of the camera lens group according to example 9, representing the curvatures of a tangential plane and the curvatures of a sagittal plane. FIG. 18C illustrates a distortion curve of the camera lens group according to example 9, representing the amounts of distortion corresponding to different field-of-views. FIG. 18D illustrates a lateral color curve of the camera lens group according to example 9, representing the deviations of different image heights on an imaging plane after light passes through the lens assembly. It can be seen from FIG. 18A to FIG. 18D that the camera lens group provided in example 9 may achieve good image quality.

(121) In view of the above, examples 1 to 9 respectively satisfy the relationship shown in Table 19.

(122) TABLE-US-00019

TABLE 19	Example	Condition	1	2	3	4	5	6	7	8	9	$f \cdot \tan(\text{Semi-FOV})$
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(mm) 4.14 3.36 2.87 3.79 2.75 4.27 3.43 4.12 2.93 DT52/DT11 4.08 3.71 3.56 3.80 3.47 4.43 3.26 4.68 3.76 TTL/ImgH 1.40 1.33 1.54 1.40 1.46 1.44 1.51 1.36 1.40 f1/R1 3.56 2.66 3.50 3.59 2.77 1.35 5.75 1.83 2.53 R5/R10 2.99 3.46 2.65 3.45 4.27 2.73 2.29 3.65 5.96 CT4/T34 1.85 1.74 1.76 2.11 2.10 3.11 3.13 2.45 1.34 (SAG42 + SAG51)/ 2.76 3.01 2.82 2.68 2.74 1.40 2.14 1.45 2.12 (SAG42 – SAG51) (ET4 + ET5)/ 2.03 2.55 1.85 1.96 2.10 3.46 2.58 3.12 2.32 (ET5 – ET4) CT5/CT3 2.13 1.05 2.13 1.82 0.96 1.53 1.65 1.53 1.19 f/f4 1.90 2.25 1.83 1.67 2.26 1.60 1.36 1.47 2.35

(123) The present disclosure further provides an imaging apparatus, having an electronic photosensitive element which may be a photosensitive Charge-Coupled Device (CCD) or a Complementary Metal-Oxide Semiconductor (CMOS). The imaging apparatus may be an independent imaging device, such as a digital camera, or may be an imaging module integrated in a mobile electronic device, such as a mobile phone. The imaging apparatus is equipped with the camera lens group described above.

(124) The foregoing is only a description of the preferred examples of the present disclosure and the applied technical principles. It should be appreciated by those skilled in the art that the inventive scope of the present disclosure is not limited to the technical solutions formed by the particular combinations of the above technical features. The inventive scope should also cover other technical solutions formed by any combinations of the above technical features or equivalent features thereof without departing from the concept of the invention, such as, technical solutions formed by replacing the features as disclosed in the present disclosure with (but not limited to), technical features with similar functions.

Claims

1. A camera lens group, sequentially from an object side to an image side of the camera lens group along an optical axis, comprising: a stop; a first lens having positive refractive power and a convex object-side surface; a second lens having positive refractive power; a third lens having negative refractive power, a convex object-side surface and a concave image-side surface; a fourth lens having positive refractive power, a concave object-side surface and a convex image-side surface; and a fifth lens having negative refractive power, a convex object-side surface and a concave image-side surface, wherein $2.00 \text{ mm} < \tan.\text{sup.2}(\text{Semi-FOV}) \cdot f < 5.00 \text{ mm}$, and $2.12 \leq (\text{SAG42} + \text{SAG51}) / (\text{SAG42} - \text{SAG51}) < 3.50$, where Semi-FOV is half of a maximum field-of-view of the camera lens group, f is a total effective focal length of the camera lens group, SAG42 is a distance measured along the optical axis from an intersection of the image-side surface of the fourth lens and the optical axis to a location projected from a vertex of an effective radius of the image-side surface of the fourth lens on the optical axis, and SAG51 is a distance measured along the optical axis from an intersection of the object-side surface of the fifth lens and the optical axis to a location projected from a vertex of an effective radius of the object-side surface of the fifth lens on the optical axis; $1.00 < f1/R1 < 6.00$, where $f1$ is an effective focal length of the first lens, and $R1$ is a radius of curvature of the object-side surface of the first lens; and $1.00 < f/f4 < 3.00$, where $f4$ is an effective focal length of the fourth lens.
2. The camera lens group according to claim 1, wherein $\text{TTL}/\text{ImgH} < 1.55$, where TTL is a distance along the optical axis from the object-side surface of the first lens to an imaging plane of the camera lens group, and ImgH is half of a diagonal length of an effective pixel area on the imaging plane of the camera lens group.
3. The camera lens group according to claim 1, wherein $2.00 < R5/R10 < 6.00$, where $R5$ is a radius of curvature of the object-side surface of the third lens, and $R10$ is a radius of curvature of the image-side surface of the fifth lens.
4. The camera lens group according to claim 1, wherein $1.00 < \text{CT4}/\text{T34} < 4.00$, where CT4 is a center thickness of the fourth lens along the optical axis, and T34 is a spaced interval between the

third lens and the fourth lens along the optical axis.

5. The camera lens group according to claim 1, wherein $1.00 < (ET4 + ET5) / (ET5 - ET4) < 4.00$, where ET4 is an edge thickness of the fourth lens, and ET5 is an edge thickness of the fifth lens.

6. The camera lens group according to claim 1, wherein $0.50 < CT5 / CT3 < 2.50$, where CT5 is a center thickness of the fifth lens along the optical axis, and CT3 is a center thickness of the third lens along the optical axis.

7. The camera lens group according to claim 1, wherein $3.00 < DT52 / DT11 < 3.00$, where DT11 is a maximum effective radius of the object-side surface of the first lens, and DT52 is a maximum effective radius of the image-side surface of the fifth lens.

8. A camera lens group, sequentially from an object side to an image side of the camera lens group along an optical axis, comprising: a stop; a first lens having positive refractive power and a convex object-side surface; a second lens having positive refractive power; a third lens having negative refractive power, a convex object-side surface and a concave image-side surface; a fourth lens having positive refractive power, a concave object-side surface and a convex image-side surface; and a fifth lens having negative refractive power, a convex object-side surface and a concave image-side surface, wherein $2.00 \text{ mm} < \tan.\text{sup.2}(\text{Semi-FOV}) * f < 5.00 \text{ mm}$, and

$1.00 < (ET4 + ET5) / (ET5 - ET4) < 4.00$, where Semi-FOV is half of a maximum field-of-view of the camera lens group, f is a total effective focal length of the camera lens group, ET4 is an edge thickness of the fourth lens, and ET5 is an edge thickness of the fifth lens; $1.00 < f1 / R1 < 6.00$, where f1 is an effective focal length of the first lens, and R1 is a radius of curvature of the object-side surface of the first lens; and $1.00 < f / f4 < 3.00$, where f4 is an effective focal length of the fourth lens.

9. The camera lens group according to claim 8, wherein $3.00 < DT52 / DT11 < 5.00$, where DT11 is a maximum effective radius of the object-side surface of the first lens, and DT52 is a maximum effective radius of the image-side surface of the fifth lens.

10. The camera lens group according to claim 9, wherein $1.00 < (SAG42 + SAG51) / (SAG42 - SAG51) < 3.50$, where SAG42 is a distance measured along the optical axis from an intersection of the image-side surface of the fourth lens and the optical axis to a location projected from a vertex of an effective radius of the image-side surface of the fourth lens on the optical axis, and SAG51 is a distance measured along the optical axis from an intersection of the object-side surface of the fifth lens and the optical axis to a location projected from a vertex of an effective radius of the object-side surface of the fifth lens on the optical axis.

11. The camera lens group according to claim 8, wherein $TTL / \text{ImgH} < 1.55$, where TTL is a distance along the optical axis from the object-side surface of the first lens to an imaging plane of the camera lens group, and ImgH is half of a diagonal length of an effective pixel area on the imaging plane of the camera lens group.

12. The camera lens group according to claim 8, wherein $2.00 < R5 / R10 < 6.00$, where R5 is a radius of curvature of the object-side surface of the third lens, and R10 is a radius of curvature of the image-side surface of the fifth lens.

13. The camera lens group according to claim 8, wherein $1.00 < CT4 / T34 < 4.00$, where CT4 is a center thickness of the fourth lens along the optical axis, and T34 is a spaced interval between the third lens and the fourth lens along the optical axis.

14. The camera lens group according to claim 8, wherein $0.50 < CT5 / CT3 < 2.50$, where CT5 is a center thickness of the fifth lens along the optical axis, and CT3 is a center thickness of the third lens along the optical axis.
